Product Requirements Document: Tuck Voter Companion

A web app that turns a voter's address into a personal voting plan and an interactive sample ballot, powered by the Google Civic Information API.

Field	Value
Product	Tuck Voter Companion (working name)
Document status	Draft v1.0
Date	November 2, 2026
Owner	Product owner (TBD)
Target release	MVP live at least 8 weeks before the next major US election date

1. Problem

Voters often don't know where their polling place is, when early voting starts, whether they can drop off a ballot, or what races and measures will be on their ballot. The information exists, but it is spread across state and county websites with inconsistent formats. Down-ballot races (school board, judges, local measures) get especially little attention, so many voters skip them or guess.

2. Goals and non-goals

Goals

- Give any US voter a complete, personal voting plan from one address entry, in under 10 seconds.
- Show the voter's full ballot with clear candidate and measure information, including local races.
- Let voters mark choices privately and take a printable or phone-friendly cheat sheet to the polls.
- Point voters to official sources for registration, absentee voting and election rules.

Non-goals (for the MVP)

- Recommending or endorsing candidates or positions. The product stays strictly nonpartisan.
- Voter registration inside the app (we link to official state sites instead).
- Elected officials lookup ("who represents me"); planned for a later phase.
- Native mobile apps; the MVP is a responsive web app.
- Coverage outside the United States.

3. Target users

Persona	Needs
First-time voter (18-24)	Plain explanations of how voting works, where to go, and what is on the ballot. Uses a phone.
Busy working adult	Fast answers: when and where, early-vote options, drop boxes near work or home. Wants reminders.
Engaged local voter	Details on down-ballot races and measures, including full text and pro/con arguments.
Recently moved voter	Confirmation of their new polling place and links to check or update registration.

4. User stories

ID	As a...	I want to...	So that...
US-1	voter	enter my home address once	I see everything that applies to me
US-2	voter	see my polling place on a map with hours	I know exactly where and when to go
US-3	voter	see early-vote sites and drop-off locations with dates	I can vote at a time that fits my schedule
US-4	voter	see every race and measure on my ballot	nothing on the ballot surprises me
US-5	voter	read candidate details and measure pro/con statements	I can make informed choices
US-6	voter	mark my choices and print or save them	I can bring them to the polls
US-7	voter	get links to official registration and absentee pages	I can fix problems before deadlines
US-8	voter	add key dates to my calendar	I don't miss early voting or election day

5. Functional requirements

Priority: **P0** = required for MVP, **P1** = should have for MVP, **P2** = later.

ID	Requirement	Priority
FR-1	Address entry with autocomplete, validation and a clear error when the address is not found.	P0
FR-2	Backend calls /elections and /voterinfo through a server-side proxy; the API key is never sent to the browser.	P0
FR-3	If several elections apply (otherElections), let the user choose one and re-query with that electionId.	P0
FR-4	Voting Plan view: election name and date, polling place with map, hours, notes, mail-only notice when mailOnly is true.	P0
FR-5	Early-vote and drop-off lists with dates and hours, sorted by distance from the address.	P0
FR-6	Sample Ballot view: contests grouped by level (federal, state, county, city, school, judicial), in ballot order.	P0
FR-7	Candidate cards: name, party, photo, website, contact and social links where available.	P0
FR-8	Measure cards: title, summary, pro and con statements, expandable full text, passage threshold.	P0
FR-9	Election office panel: state and local links (registration, check registration, absentee, rules) and contact details.	P0
FR-10	Source labels on all data, marking official vs. unofficial sources.	P0
FR-11	Empty state when no election data is available, explaining why and linking to the state election office.	P0
FR-12	Mark choices on the ballot; saved only in the user's browser; print and "save as image" cheat sheet.	P1
FR-13	Add-to-calendar (.ics) for early-vote start, drop-off deadline and election day.	P1
FR-14	Shareable link to the Voting Plan that does not contain the full address.	P1
FR-15	Spanish translation of the interface.	P1
FR-16	Plain-language AI summary of long measure text, clearly labeled, with the full official text one tap away.	P2
FR-17	Email/SMS reminders with opt-in and one-tap unsubscribe.	P2
FR-18	Who Represents Me (divisionByAddress + OpenStates + Congress.gov).	P2

6. Non-functional requirements

Area	Requirement
Performance	Results shown within 3 seconds at the 95th percentile on a 4G phone; first page load under 2 seconds.
Privacy	No accounts in the MVP. Addresses are not stored server-side or written to logs; choices stay in the browser. No third-party ad trackers.
Security	API key stored as a server secret and restricted in Google Cloud; rate limiting per IP on the proxy; HTTPS only.
Accessibility	WCAG 2.2 AA: keyboard navigation, screen-reader labels, sufficient contrast, readable at 200% zoom.
Reliability	Cache voterinfo responses briefly (e.g. 15 minutes) per address and election to absorb traffic spikes and stay within quota; 99.5% uptime in the 30 days before an election.
Neutrality	Candidates shown in official ballot order; no ranking, endorsements or editorial content.
Devices	Mobile-first responsive layout; works on current Chrome, Safari, Firefox and Edge.

7. Data and API usage

Screen	Endpoint	Key fields
Election picker	GET /elections; voterinfo.otherElections	id, name, electionDay
Voting Plan	GET /voterinfo	pollingLocations, earlyVoteSites, dropOffLocations, mailOnly, normalizedInput
Sample Ballot	GET /voterinfo	contests (office, level, district, ballotPlacement, candidates, referendum fields)
Election office	GET /voterinfo	state[].electionAdministrationBody, local_jurisdiction
All screens	GET /voterinfo	sources[] (name, official)

8. Proposed architecture

- **Frontend:** Next.js (React) with TypeScript and Tailwind CSS; a map component (e.g. Leaflet with OpenStreetMap tiles, or Google Maps).
- **Backend:** Next.js API routes acting as a thin proxy: `/api/elections` and `/api/voterinfo`. They add the key, validate input, rate-limit, cache, and normalize the response into the app's own types.
- **Cache:** in-memory for development; a managed key-value store (e.g. Redis) in production.
- **Hosting:** Vercel or a similar platform, with the API key stored as an environment secret.
- **Analytics:** privacy-friendly, cookieless page analytics only; no addresses in events.

9. Success metrics

Metric	Target (first election cycle)
Address lookups that return a Voting Plan	At least 90% during the 30 days before an election
Time from address entry to results	Under 3 seconds (95th percentile)
Users who open the Sample Ballot	At least 50% of successful lookups
Cheat sheets printed or saved	At least 15% of Sample Ballot visitors
Calendar adds or reminder opt-ins	At least 10% of successful lookups
Accessibility audit	Zero critical WCAG AA issues at launch

10. Risks and mitigations

Risk	Mitigation
Data is missing between elections or arrives late.	Clear empty state with official links; build and test with electionId=2000; launch the marketing push only once Google publishes data.
Incomplete local coverage in some states.	Show what is available, label gaps, and always link to the county election office.
Traffic spikes near election day exceed quota.	Response caching, request a quota increase early, and load-test before launch.
Wrong or outdated information.	Show sources, prefer officialOnly for critical data, add a "report a problem" link.
Perceived partisanship.	Official ballot order, no endorsements, neutral copy reviewed before launch.
Google changes or retires more of the API.	Isolate API calls behind the proxy and app-level types so another data provider can be swapped in.

11. Milestones

Phase	Scope	Duration
0. Setup	Repo, API key, proxy endpoints, test election data	Week 1
1. Voting Plan	FR-1 to FR-5, FR-9 to FR-11	Weeks 2-3
2. Sample Ballot	FR-6 to FR-8, FR-12	Weeks 4-5
3. Polish	FR-13 to FR-15, accessibility audit, load testing	Weeks 6-7
4. Launch	Production deploy, monitoring, feedback channel	Week 8
5. Later	FR-16 to FR-18	After launch

12. Open questions

- Final product name and branding.
- Which map provider to use (cost vs. features).
- Whether to require officialOnly=true for polling locations.
- Which languages to support beyond English and Spanish.
- Whether the later "Who Represents Me" phase uses free sources only or a paid provider.